

Report for generation of SPP on Si and Ti targets in water medium

Thibault J.-Y. Derrien, Jörg Krüger, and Jörn Bonse
BAM Bundesanstalt für Materialforschung und -prüfung,
Unter den Eichen 87, D-12205 Berlin, Germany
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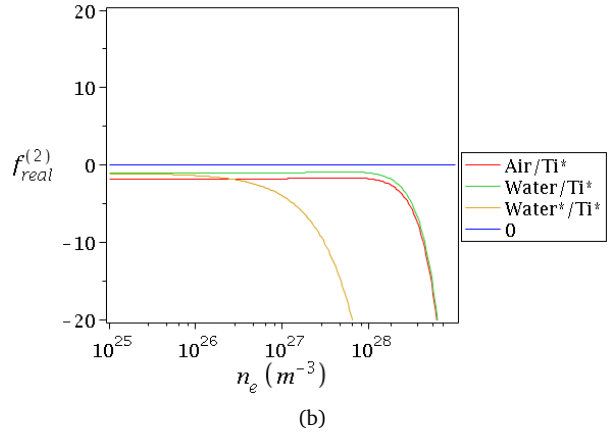
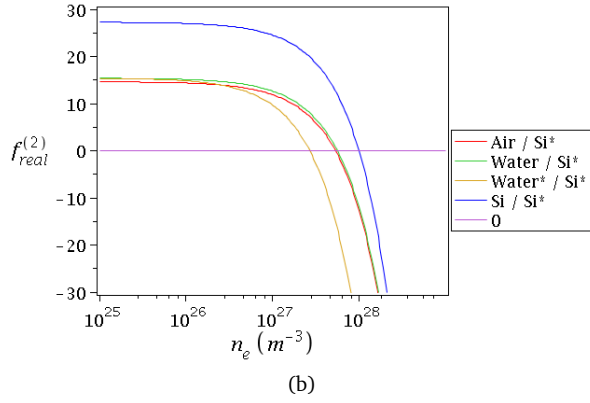
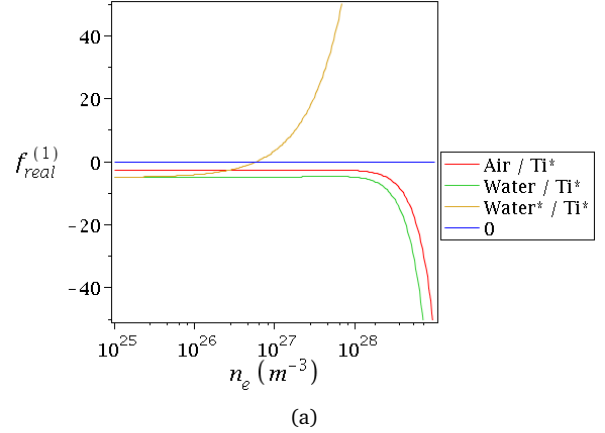
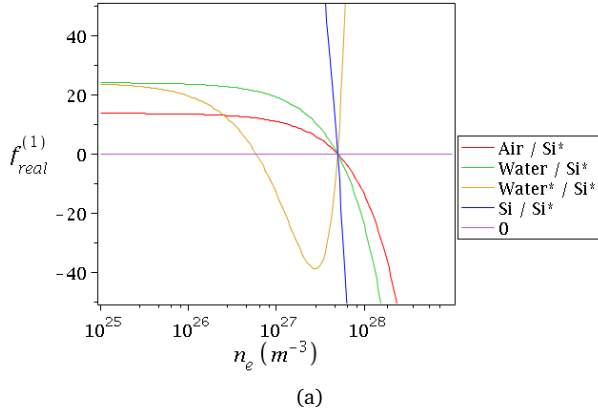


Figure 1: Conditions for SPP excitation by neglecting absorption at interface with Si in different media. (a) First condition. (b) Second condition.

Figure 2: Conditions for SPP excitation by neglecting absorption at interface with Ti in different media. (a) First condition. (b) Second condition.

I. DIELECTRIC FUNCTIONS

The dielectric function for Ti is taken from Golosov et al. [1]. For Si, we use dielectric function given by Sokolowski-Tinten and von der Linde [2].

II. SPP EXCITATION AT A SINGLE INTERFACE

A. Conditions neglecting interband absorption

Condition for SPP excitation neglecting the absorption of the external medium is given by both conditions $f_{real}^{(1)} = \Re(\varepsilon_1) \Re(\varepsilon_2) < 0$ and $f_{real}^{(2)} = \Re(\varepsilon_1) + \Re(\varepsilon_2) < 0$ [3–5] where $\varepsilon_1 \in \mathbb{R}$ and $\varepsilon_2 \in \mathbb{C}$. Figure 1 presents the values of these quantities as a function free-carrier density, at the interface with Si.

Figure 2 present the conditions at the interface with Ti.

B. Conditions considering interband absorption

Condition for SPP excitation at a single interface between media 1 and 2 is given by [6]

$$f(\varepsilon_1, \varepsilon_2) < 0 \quad (1)$$

where

$$f(\varepsilon_1, \varepsilon_2) = \Re(\varepsilon_1) \Re(\varepsilon_2) + \Im(\varepsilon_1) \Im(\varepsilon_2)$$

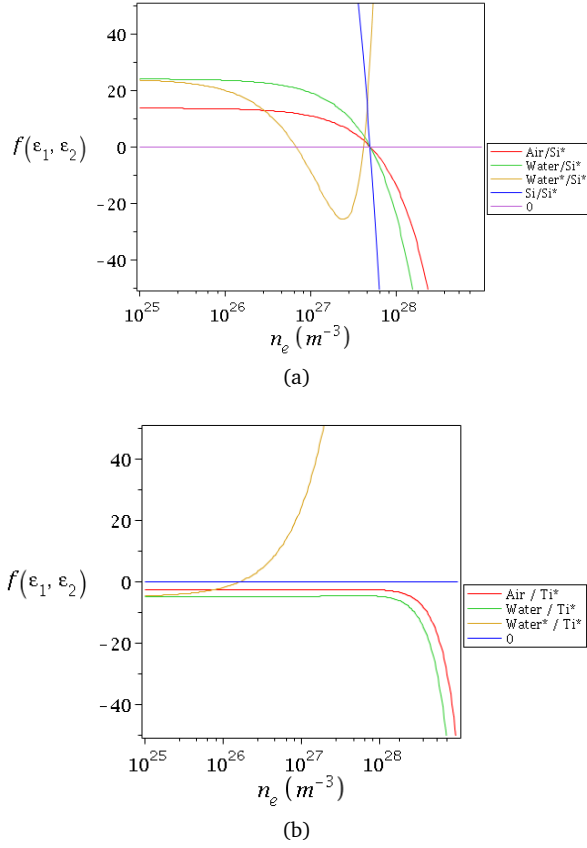


Figure 3: Validity domain for SPP condition at interfaces involving (a) Si, (b) Ti.

Figure 3 defines the possible interfaces at which SPP can be excited. Results show possible excitation at :

- air / excited Si at very high free-carrier density in Si
- water / excited Si at very high free-carrier density in Si
- excited water / excited Si in a specific range of carrier density in Si and water
- excited water / Ti while water is still a dielectric

Figure 4 presents the resulting periodicities for each case at which SPP can be excited.

- air / Si*: $\Lambda_{SPP} \sim 800$ nm.
- water / Si*: $\Lambda_{SPP} \sim 630$ nm.
- water* / Si*: $400 \text{ nm} < \Lambda_{SPP} < 1400 \text{ nm}$
- water* / Ti*: $600 \text{ nm} < \Lambda_{SPP} < 650 \text{ nm}$
- air / Ti*: $\Lambda_{SPP} \sim 800 \text{ nm}$

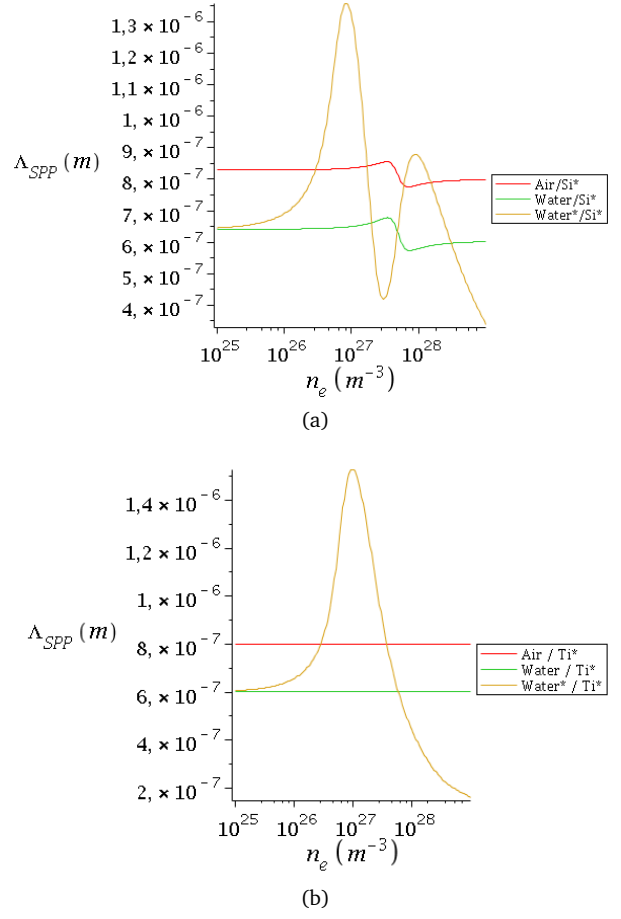


Figure 4: SPP periodicity at the interfaces (a) with Si, (b) with Ti.

III. MULTILAYER SPP CALCULATIONS

The equation for SPP excitation in a multilayer system is given by [5, 7]

$$e^{-4k_1 a} = \frac{\frac{k_1}{\varepsilon_1} + \frac{k_2}{\varepsilon_2} \frac{k_1}{\varepsilon_1} + \frac{k_3}{\varepsilon_3}}{\frac{k_1}{\varepsilon_1} - \frac{k_2}{\varepsilon_2} \frac{k_1}{\varepsilon_1} - \frac{k_3}{\varepsilon_3}} \quad (2)$$

where

$$k_i^2 = \beta^2 - k_0^2 \varepsilon_i \quad (3)$$

$k_i, i \in \{1, 2, 3\}$ is the complex wave number in medium i . β is the complex wavenumber characterizing the SPP modes in the thin layer. $\varepsilon_i, i \in \{1, 2, 3\}$ are the dielectric functions characterizing the optical properties of each medium. k_0 is the incident laser wavenumber. $t = 2a$ is the thickness of the thin layer.

The model has been verified by reproducing Prade et al. [7] results with a very good match.

In case of two identical interfaces, the SPP wave number is trivial. Performing analytical calculation over the multilayer model leads to a trivial solution, equivalent

to the case $a = 0$

$$\beta_{a=0} \in \left\{ \pm k_0 \sqrt{\varepsilon_1}, \pm k_0 \sqrt{\frac{\varepsilon_2 \varepsilon_3}{\varepsilon_2 + \varepsilon_3}} \right\}$$

which describes the SPP at the interface *air* – *Ti*. The physical set of solution is reduced to

$$\beta_{a=0} = \pm k_0 \sqrt{\frac{\varepsilon_2 \varepsilon_3}{\varepsilon_2 + \varepsilon_3}} \quad (4)$$

which well corresponds to the case of a single interface made by media II and III. However, calculating the wave number β at a trivial interface leads to a singularity that we well identified analytically. However, the degeneracy of the solution can lead to a singular solution using the numerical resolution.

IV. EXPERIMENTAL MEASUREMENTS

See Table II.

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Medium 1	Layer	Medium 2	$\Lambda_{SPP}(\text{nm})$	$a = \frac{h}{2}(\text{nm})$	$n_e(\text{cm}^{-3})$	ν_1^{-1}	ν_2^{-1}	ν_3^{-1}	$T_e(\text{Ti}^*)$	grid [$N_{n_e} \times N_a$]
air	Si*	Si	10 – 100	0 – 3	$6 \times 10^{21} - 10^{23}$	-	1.1 fs	-	-	20×20
Water	Water*	Si*	10 – 100	< 5	$10^{21} - 10^{22}$	-	1.0 fs	1.1 fs	-	10×10
Water	Water*	Si	10 – 100	< 5	$10^{21} - 10^{23}$	-	1.0 fs	-	-	10×10
Water	Si*	Si	10 – 100	< 3.5	$5 \times 10^{21} - 10^{23}$	-	1.1 fs	-	-	10×10
air	Ti*	Ti	800	-	-	1.0 fs [8]	1.0 fs	1.7 fs [9]	10 kK	20×20
water*	Ti*	Ti	200	20 – 100	$5 \times 10^{21} - 5 \times 10^{22}$				10 kK	5×5
water	Ti*	Ti	600	1 – 100	$5 \times 10^{21} - 10^{23}$				10 kK	5×5
water	water*	Ti	540 – 610	1 – 50	$2 \times 10^{21} - 1 \times 10^{23}$				-	10×5
water	water*	Ti	350 – 600	1 – 50	$2 \times 10^{21} - 1 \times 10^{23}$	1.0 fs	1.0 fs	1.0 fs	-	10×10
water	water*	Ti*	500 – 800	1 – 50	$6 \times 10^{19} - 1 \times 10^{23}$				10 kK	5×10
water	water*	Ti*	500 – 600	1 – 50	$2 \times 10^{21} - 1 \times 10^{23}$				50 kK	5×5
water	TiO ₂ *	Ti	300 – 600	1 – 50	$2.4 \times 10^{19} - 7 \times 10^{20}$				1.0 fs	10×10

Table I: SPP period as a function of the stack layered structure

Material	Medium	N	Period	Experimental match
Si	air	100	555 – 585 nm	Single SPP with blueshift
Si	water	100	80 – 130 nm	Multilayer SPP (several cases)
Ti	air	1000	80 nm	Multilayer SPP with air/TiO ₂ */Ti
Ti	water	1000	70 – 170 nm	Multilayer SPP Water*/Ti*/Ti

Table II: Experimental periods of LSFL ripples in air and water.